



Indian Association for the Cultivation of Science  
(Deemed to be University under *de novo* Category)  
Integrated Bachelor's-Master's Program

Mid-Semester Examination-Autumn 2025

Subject: Chemical Reactivity

Full Marks: 25

Subject Code(s): CHS 2101

Time Allotted: 2 h

Section-A (14 Marks)

1. Answer the following questions (*Any three*).

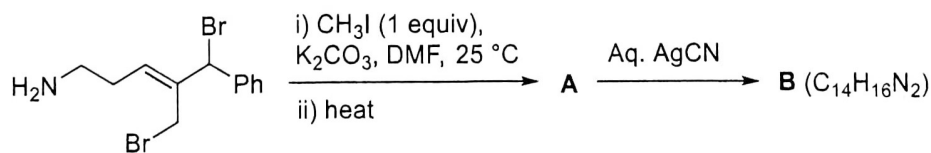
[3x1.5 = 4.5]

- (a) The rate of hydrolysis of  $\text{EtSCH}_2\text{CH}_2\text{Cl}$  is  $10^4$  times faster than that of  $\text{EtOCH}_2\text{CH}_2\text{Cl}$  under a given condition. Explain the observation with a proper reaction mechanism.
- (b) Although tosylate is a good leaving group, the given compound does not undergo substitution reactions efficiently. Explain.



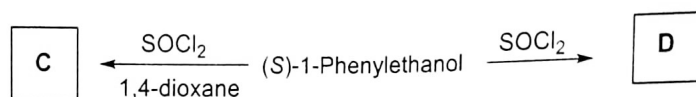
- (c) The conversion of  $\text{PhCOCl}$  to  $\text{PhCO}_2\text{CH}_2\text{Ph}$  occurs at a significantly faster rate in the presence of a base and a catalytic amount of 4-dimethylaminopyridine (DMAP). Explain the role of DMAP in the reaction and draw the key reaction intermediate.
- (d) Often sigma-bonds rotate freely, but pi-bonds do not. Explain with a suitable example using the bonding orbital representation.
- (e) Describe the  $\text{E1cB}$  reaction with a proper example.

2. (a) Predict the structure of product **A** (major product) and products **B** (all possible isomers) for the below given transformations, and justify your answer with proper reaction mechanism. [3]



Or

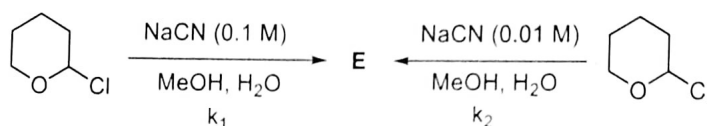
Predict the structures of the expected products **C** and **D** of the following reactions with proper stereochemistry, reaction mechanisms, and justification. How are these products (**C** and **D**) related stereochemically? [3]



(b) Draw the most stable conformer of all *trans*-hexachlorocyclohexane and explain why it undergoes E2 reaction at a very slow rate in a given condition. [1.5]

3. (a) Nucleophilic aromatic substitution reaction is difficult but not impossible. Justify with a suitable example. [1.5]

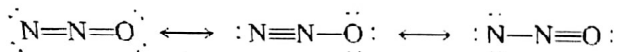
(b) Predict the structure of **E**. How are the rates ( $k_1$  and  $k_2$ ) of the following reactions related? Explain. [2]



(c) Explain the stereoelectronic requirement of an E2 reaction. [1.5]

### Section-B (11 Marks)

4. (a) Nitrous oxide ( $\text{N}_2\text{O}$ ) has three possible Lewis structures: [2]



Given the following bond lengths,

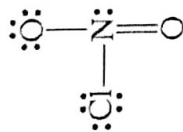
|     |        |     |        |
|-----|--------|-----|--------|
| N—N | 167 pm | N=O | 115 pm |
| N=N | 120 pm | N—O | 147 pm |
| N≡N | 110 pm |     |        |

rationalize the observations that the N-N bond length in  $\text{N}_2\text{O}$  is 112 pm and that the N-O bond length is 119 pm. Assign formal charges to the resonance structures for  $\text{N}_2\text{O}$ . Can you eliminate any of the resonance structures based on formal charges? Is this consistent with observation?

Or

Write down the Lewis dot formula of the oxoacid anion,  $\text{IO}_4^-$ . What is the oxidation state of iodine in the oxo acid? What is the basicity of this oxoacid? [2]

- (b) What changes must be made to the following Lewis structure to correct it? [1.5]



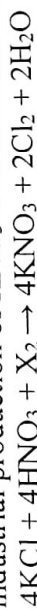
5. (a) What will be the ideal geometry for  $\text{AB}_5$ -type molecules? How does this ideal geometry deviate for  $\text{AB}_3\text{E}_2$ -type molecules (E stands for lone pair)? [2]

Or

Each of the compounds  $\text{Pt}(\text{NH}_3)_6\text{Cl}_4$ ,  $\text{Cr}(\text{NH}_3)_6\text{Cl}_3$ ,  $\text{Co}(\text{NH}_3)_4\text{Cl}_3$ , and  $\text{K}_2\text{PtCl}_6$  is separately dissolved in water to make 0.01 M solutions. Rank them in order of their increasing conductivity in a solution. [2]

- (b) A compound of arsenic and fluorine is a gas. A sample weighing 0.100 g occupies 14.2 mL at  $23^\circ\text{C}$  and 765 mmHg. What is the molecular mass of the compound? Write the Lewis formula for the molecule, showing a formal charge on each atom. What would be the geometry of the molecule? (Given:  $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$ , atomic mass of arsenic = 74.922 u) [2]

6. (a) Potassium nitrate is used in the production of fireworks, explosives, and matches. It is also used in curing foods and to modify the burning properties of tobacco. The reaction for the industrial production of  $\text{KNO}_3$  is summarized below.



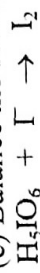
What is X in this reaction?

[1.5]

Or

- (b) How many electrons are required to reduce  $\text{N}_2$  to  $\text{NH}_3$ ? Show the half-reaction. [1.5]

- (c) Balance the redox reaction.



Will the reaction be more or less favorable if the pH of the reaction is decreased? [2]